MM : 720

TEST - 1

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer:

1. If dimensional formula of physical quantity CV^2 is $[M^aL^bT^c]$, then the value of [redacted] will be (Here V is voltage and C is capacitance)

(1) 0 (2) 2

(3) 1 (4) $\frac{1}{2}$

2. If the pitch of a screw gauge is [redacted] mm and there are [redacted] divisions in its circular scale, then the least count of the screw gauge will be

(1) 0.01 mm (2) 0.02 mm

(3) 0.03 mm (4) 0.04 mm

3. A particle is moving in a circular path, such that at any instant its angular position (θ) can be given as $\theta = \frac{5t^3}{30} - \frac{t^2}{2}$; here θ is in radian and t is in second. The angular acceleration of the particle at t = 10 second is (in rad/s^2)

(1) 2 (2) 4

(3) 6 (4) 9

4. The velocity of a particle moving in x - y plane is given as $\vec{v} = (-x\hat{i} + 2y\hat{j})$ m/s. The magnitude of acceleration of particle at point (1, 2) m is

(1) $\sqrt{65}$ m/s^2 (2) $\sqrt{33}$ m/s^2

(3) $\sqrt{26}$ m/s^2 (4) $\sqrt{45}$ m/s^2

5. Two projectiles are projected from the ground with same initial speed at the angles [redacted] and [redacted] from horizontal. The ratio of their horizontal ranges is 1 : x. The value of x is

(1) 1 (2) 2

(3) 3 (4) $\sqrt{3}$

6. The velocity of a moving particle along x-axis is given by $v = (2t)$ m/s. If position (x) of the particle is origin at time t = 0, then position of the particle at t = 2 s will be

(1) x = 2 m (2) x = 4 m

(3) x = 8 m (4) x = 1 m

7. Some physical quantities are given in list-I and their dimensional formulae are given in list-II. Match the list-I with list-II and choose the correct option.

	List-I (Physical quantity)		List-II (Dimensional formulae)
a.	Angular frequency	(i)	$[ML^2T^{-2}]$
b.	Torque	(ii)	$[M^0L^0T^{-1}]$
c.	Electric potential	(iii)	$[ML^2T^{-1}]$
d.	Planck's constant	(iv)	$[ML^2T^{-3}A^{-1}]$

(1) a(ii), b(iii), c(iv), d(i)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(i), d(iv)

8. Two cars A and B are moving in the same direction along straight road with constant speed [redacted] km/h and [redacted] km/h respectively. Car B is moving ahead of car A. A person in car A throws a stone with speed v (relative to car A) towards car B, so that it hits the car B with speed of 10 m/s. The value of v in km/h is

(1) 18 (2) 28

(3) 36 (4) 16

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9. Co-ordinates of a projectile moving in x-y plane as a function of time (t) are given as $x = 20t$ and $y = (50t - 5t^2)$, where x is along horizontal and y is along vertical. The angle made by the velocity of projectile with horizontal at $t = 2$ second is α. Then $\tan\alpha$ equals

(1) $\frac{2}{3}$ (2) $\frac{3}{2}$

(3) $\frac{4}{3}$ (4) $\frac{3}{4}$

10. Two projectiles are fired from ground with same initial speed from same point at angles $(45° + \alpha)$ and $(45° - \alpha)$ with horizontal direction respectively, then

(1) Their horizontal ranges are same

(2) Their average velocities are different for complete flight

(3) Their average velocities are same for complete flight

(4) Both (1) and (2)

11. In an experiment to determine the resistance of a given wire using Ohm's law, the voltmeter and ammeter readings are 10 V and 5 A respectively. The least count of voltmeter and ammeter are 100 mV and 200 mA respectively. The calculated maximum error in resistance is

(1) 0.1 Ω (2) 0.2 Ω

(3) 0.3 Ω (4) 0.4 Ω

12. A particle starts from rest with constant angular acceleration and moves on a circle. In the first 3 seconds it covers angular displacement θ_1 and in next 3 second it covers angular displacement θ_2. The ratio $\frac{\theta_2}{3\theta_1}$ is

(1) 6 (2) 3

(3) 1 (4) $\frac{1}{3}$

13. A gas balloon is going up with constant velocity of 10 m/s. When this balloon reached a height of 75 m from ground, a stone is dropped from it and balloon keeps moving up with the same velocity. The height of balloon when the stone hits the ground will be ($g = 10$ m/s^2)

(1) 75 m (2) 350 m

(3) 135 m (4) 125 m

14. A new unit of length is chosen to be α and it is numerically equal to speed of light in vacuum. What is the distance between Mars and Earth in terms of α units if light takes 12 minutes 30 seconds to cover this distance?

(1) 200α (2) 400α

(3) 750α (4) 500α

15. A person travelling on a straight line (without turning back) moves with uniform velocity v_1 for a distance $\frac{x}{2}$ and with a uniform velocity v_2 for the next $\frac{3x}{2}$ distance. The average velocity of the person for complete motion is 20 m/s. If v_1 is 10 m/s, then v_2 equals

(1) 10 m/s (2) 20 m/s

(3) 30 m/s (4) 40 m/s

16. The total number of significant zeros in the measurement 0.01020×10^4 m is

(1) 2 (2) 3

(3) 4 (4) 8

17. Two physical quantities A and B have different dimensions. Which of the following mathematical operation is certainly not meaningful?

(1) $Z = A \times B$ (2) $Z = \frac{A \times B}{A^2 + B}$

(3) $Z = \frac{A^2 B}{B^2 + A}$ (4) $Z = \frac{AB^2}{A + B}$

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18. Precision in measurement depends upon
(1) Zero error
(2) Parallax
(3) Least count of instrument
(4) All of these

19. Read the following statements carefully and choose the correct option.

Statement A: Random error can be minimised by taking more number of readings.

Statement B: Number of reliable digits plus one uncertain digit are known as significant figures in a measurement.

(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

20. Considering density (ρ), frequency (f) and velocity (v) as fundamental quantities in a system of units, then dimensional formula of force will be
(1) $[\rho f^{-2} v^4]$ (2) $[\rho f v^3]$
(3) $[\rho f^{-2} v^3]$ (4) $[\rho f^{-2} v^2]$

21. The length of a rectangle is 2·1 m and its width is 1·62 m. Then its area will be
(1) 3.402 m^2 (2) 3.4 m^2
(3) 3.40 m^2 (4) 3 m^2

22. A new system of unit is chosen in which the unit of mass is taken as 10 kg, unit of length is taken as 1 m and unit of time is taken as 10 second. 1 joule in this new system is numerically equal to
(1) 1 unit
(2) 1000 unit
(3) 10 unit
(4) $\frac{1}{10}$ unit

23. A particle moves along a quadrant of circle of radius 4 m with a constant speed 2 m/s. The magnitude of average acceleration in this interval will be (in m/s^2)
(1) Zero (2) $2\sqrt{2}\pi$
(3) $\frac{2\sqrt{2}}{\pi}$ (4) $\frac{2}{\pi}$

24. A ground-to-ground projectile can have the same range for two angles of projection. If times of flight in both the above cases are t_1 and t_2, then speed of projection of the projectile is
(1) $\frac{1}{2}g(t_1+t_2)$ (2) $\frac{1}{4}g(t_1+t_2)$
(3) $\frac{1}{2}g\left(t_1^2+t_2^2\right)^{\frac{1}{2}}$ (4) $\frac{1}{2}g\left[t_1^2+t_2^2\right]$

25. Read the following statements carefully and choose the correct option (symbol have their usual meaning).

Statement A: $\frac{d\vec{v}}{dt}$ is net acceleration of particle.

Statement B: $\left|\frac{d\vec{v}}{dt}\right|$ is magnitude of net acceleration of particle.

Statement C: $\frac{d|\vec{v}|}{dt}$ is tangential acceleration of particle.

(1) All statements A, B and C are correct
(2) Only statements A and B are correct
(3) Only statements B and C are correct
(4) Only statement B is correct

26. A helicopter is flying horizontally with speed 360 km/h at an altitude 2 km. It drops an object which hits the ground after sometime. Displacement of object as seen from the helicopter would be ($g = 10\ m/s^2$)
(1) $2\sqrt{2}$ km (2) 2 km
(3) $2\sqrt{3}$ km (4) 3 km

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27. An open trolley car of length 25 m, is moving with constant speed 5 m/s on a straight path. A boy moves on trolley from rear end to front end and then returns to rear end in 10 seconds. During complete journey of boy

(1) Displacement of boy w.r.t. ground is 500 m

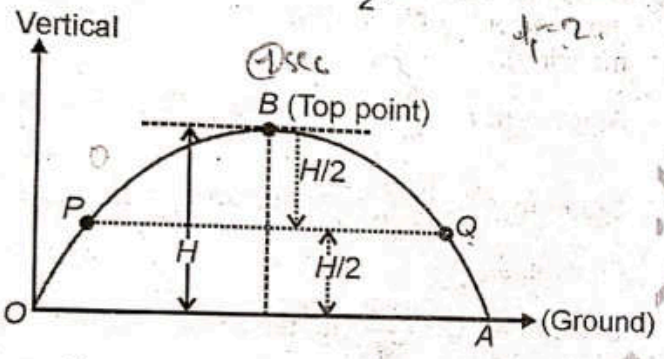
(2) Displacement of trolley is 500 m

(3) Displacement of boy w.r.t. trolley is zero

(4) All of these

28. Trajectory of a particle projected from point '*O*' on ground is shown in the figure.

If maximum height attained is *H* and time taken by the particle to reach from *O* to *A* is 2 seconds, then time taken by the particle to reach from *P* to *Q* (symmetrically at height $\frac{H}{2}$) will be



(1) $\frac{1}{2}$ s

(2) $\frac{3}{4}$ s

(3) $\sqrt{2}$ s

(4) $\frac{1}{\sqrt{2}}$ s

29. Maximum and minimum value of the magnitude of addition of two vectors are 5 unit and 3 unit respectively. Then product of the magnitudes of these two vectors is (in proper unit)

(1) 8

(2) 16

(3) 4

(4) 5

30. In ground-to-ground oblique projectile motion, whe the projectile reaches the topmost point of it trajectory, then at this instant

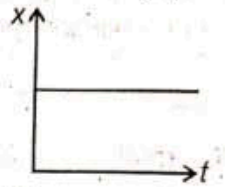
(1) Its speed is zero

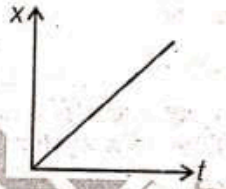
(2) Its acceleration is *g*

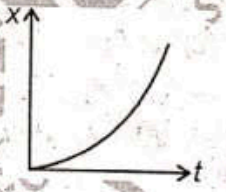
(3) Its kinetic energy is minimum

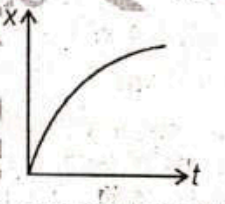
(4) Both (2) and (3)

31. Consider a car undergoing uniform motion. Th position-time (*x*-*t*) graph of the car will be bes represented by

(1)


(2)


(3)


(4)


32. The angular velocity of a particle moving in a circular orbit with 100 revolutions per minute is nearly

(1) 1.66 rad/s

(2) 10.47 rad/s

(3) 1.47 rad/s

(4) 60 rad/s

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33. A stone is dropped from the top of a tower of height 19.6 m above the ground. The velocity of the stone just before it hits the ground, is (take $g = 9.8\ m/s^2$)
(1) 4.9 m/s
(2) 9.8 m/s
(3) 19.6 m/s
(4) 39.2 m/s

34. Read the **Assertion (A)** and **Reason (R)** carefully to mark the correct option out of the options given below.

Assertion (A): If the speed of a body is constant, the body cannot have a path other than a circular or straight line path.

Reason (R): It is not possible for body to have a constant speed in an accelerated motion.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

35. In uniform circular motion of a particle, its
(1) Acceleration is zero
(2) Velocity is const.
(3) Angular displacement (θ) varies linearly with time
(4) Both (2) and (3)

36. A boy moves 12 m along east and then he moves 16 m along north and stops. Net displacement of the boy during his journey is
(1) 28 m
(2) 4 m
(3) 20 m
(4) 40 m

37. One centimetre on the main scale of vernier calliper is divided equally into 10 equal parts. If 10 divisions of vernier scale coincide with 9 divisions of the main scale, then least count of the vernier calliper is
(1) 0.1 cm
(2) 0.5 cm
(3) 0.05 cm
(4) 0.01 cm

38. Read the following statements carefully and choose the correct option

Statement A: Mass, time and velocity cannot be fundamental quantities in any system of units.

Statement B: Dimensionless quantity never has units.

Statement C: Unitless quantity does not exist.

(1) All statements A, B, and C are incorrect
(2) Only statements A and B are correct
(3) Only statement B and C are correct
(4) Only statement B is correct

39. Force acting on a particle is given by $F = Pt^{-1} + Qt$, where t is time while P and Q are dimensional constants. The unit of P is same as that of
(1) Acceleration
(2) Momentum
(3) Velocity
(4) Displacement

40. A body starts from rest and covers a certain distance in time t. If its acceleration is constant, then it will cover second half of that distance in time
(1) $\frac{t}{\sqrt{2}}$
(2) $\frac{t}{\sqrt{3}}$
(3) $t\left(1-\frac{1}{\sqrt{2}}\right)$
(4) $t\left(1-\frac{1}{\sqrt{3}}\right)$

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41. Speed of river is 4 km/h and speed of swimmer with respect to river is 3 km/h. In which direction swimmer must swim so that he will reach a point directly opposite to the starting point?

(1) At 127° with direction of river flow

(2) At 90° with direction of river flow

(3) At 143° with direction of river flow

(4) Swimmer will never reach a point directly opposite to the starting point

42. When a person walks on a straight road with speed 10 km/h, rain appears to fall vertically downward (to him). As he stops, the rain appears to fall at angle 30° with vertical. The speed of rain with respect to him (when he was walking), is

(1) 20 km/h

(2) $10\sqrt{3}$ km/h

(3) 10 km/h

(4) $20\sqrt{3}$ km/h

43. A body is projected from ground with speed 25 m s^{-1} at an angle 30° with horizontal. Then at what time (before it hits the ground) its velocity will be perpendicular to its initial direction of projection?

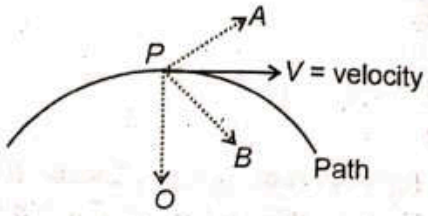
(1) 2 s

(2) 1.7 s

(3) 3 s

(4) Such case is not possible

44. A body moves on a circular path having centre O as shown in the figure. At any time instant particle is at point P moving with velocity V, then



(1) Net acceleration of particle at P may be towards centre O as $\overrightarrow{PO}$

(2) Net acceleration of particle at P may be outside the curvature as $\overrightarrow{PA}$

(3) Net acceleration of particle at P may be inside the curvature as $\overrightarrow{PB}$

(4) Both (1) and (3)

45. For a particle moving on a circle of radius (r), its speed (v) as a function of time (t) is given by $v = kt$, where k is positive constant. Angle between acceleration and velocity at any time instant t is

(1) $\tan^{-1}\left(\frac{kt^2}{r}\right)$

(2) $\cot^{-1}\left(\frac{kt^2}{r}\right)$

(3) $\sin^{-1}\left(\frac{kt^2}{r}\right)$

(4) $\cos^{-1}\left(\frac{kt^2}{r}\right)$

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[CHEMISTRY]

46. Which of the following set of quantum numbers is not possible?

(1) $n = 1,\ \ell = 0,\ m_\ell = 0,\ m_s = +\frac{1}{2}$

(2) $n = 4,\ \ell = 3,\ m_\ell = -2,\ m_s = -\frac{1}{2}$

(3) $n = 5,\ \ell = 1,\ m_\ell = -2,\ m_s = +\frac{1}{2}$

(4) $n = 2,\ \ell = 1,\ m_\ell = 0,\ m_s = -\frac{1}{2}$

47. The number of emission spectral lines observed in visible spectrum when the excited electron of a sample of H atoms in 4th excited state drops to ground state is

(1) 3

(2) 10

(3) 4

(4) 5

48. Given below are two statements:

Statement I: Heisenberg uncertainty principle rules out existence of definite paths or trajectories of electrons.

Statement II: During the photoelectric effect, the kinetic energy of ejected electrons increase with the increase of frequency of the light used.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are false

(2) Both statement I and statement II are true

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

49. Which among the following ions has maximum number of unpaired electrons?

(1) Fe^{2+} (2) Mn^{2+}

(3) Cr^{3+} (4) Co^{2+}

50. Match column-I with column-II.

	Column-I (Property about subshell)		Column-II (Orbital angular momentum)
a.	Spherically symmetrical	(i)	$\sqrt{6}\frac{h}{2\pi}$
b.	For this subshell to exist, lowest value of n is 4	(ii)	$\sqrt{2}\frac{h}{2\pi}$
c.	Number of lobes = 4	(iii)	$\sqrt{12}\frac{h}{2\pi}$
d.	Electron density present only along z-axis	(iv)	0

Choose the correct option.

(1) a(iv), b(iii), c(i), d(ii) (2) a(i), b(ii), c(iii), d(iv)

(3) a(iv), b(i), c(iii), d(ii) (4) a(ii), b(iii), c(iv), d(i)

51. The energy required to excite 1.12 L of hydrogen gas at 1 atm and 273.15 K from ground state to the second excited state of atomic hydrogen is

(The energy for the dissociation of H – H bond is 436 kJ/mol)

(1) 70.3 kJ

(2) 98.4 kJ

(3) 138.4 kJ

(4) 100.3 kJ

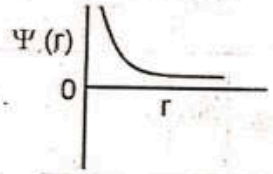
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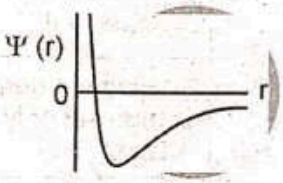
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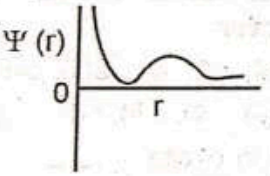
52. Which of the following transitions of the electron in the hydrogen atom emits maximum energy photon?
(1) $n_2 = 2$ to $n_1 = 1$
(2) $n_2 = 5$ to $n_1 = 4$
(3) $n_2 = 3$ to $n_1 = 2$
(4) $n_2 = 4$ to $n_1 = 3$

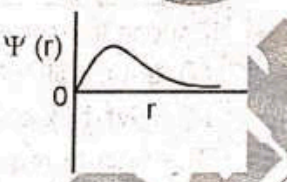
53. The ratio of the energy of a photon of 300 Å wavelength radiation to that of 900 Å radiation is
(1) 2 : 1
(2) 1 : 3
(3) 3 : 1
(4) 1 : 4

54. The graph is plotted between orbital wave function ($\Psi(r)$) and distance r of the electron from the nucleus. Which among the following is correctly plotted for 2*s* orbital?

(1) 

(2) 

(3) 

(4) 

55. The correct outermost electronic configuration of element with atomic number 46 is
(1) $4d^8\ 5s^2$ (2) $4d^{10}\ 5s^0$
(3) $4d^9\ 5s^1$ (4) $4d^9\ 5s^0$

56. The correct decreasing order of energy of 2s orbital for the given elements is
(1) K > Na > Li > H (2) H > Li > Na > K
(3) Li > H > Na > K (4) K > Li > H > Na

57. The total number of electrons that can be accommodated in 5g subshell is
(1) 18 (2) 25
(3) 9 (4) 14

58. Match column-I with column-II. (Z-atomic number)

	Column-I		Column-II
a.	Radius of nth orbit $\propto \frac{n^x}{Z^y}$	(i)	x = y = –1
b.	Velocity of electron in n^{th} orbit $\propto \frac{n^x}{Z^y}$	(ii)	x = y = –2
c.	Magnitude of energy of an electron in n^{th} orbit $\propto \frac{n^x}{Z^y}$	(iii)	x = 1, y = 0
d.	Angular momentum in n^{th} orbit $\propto \frac{(n)^x}{(Z)^y}$	(iv)	x = 2, y = 1

Choose the correct option.
(1) a(i), b(ii), c(iii), d(iv)
(2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(iv), c(i), d(ii)
(4) a(iv), b(i), c(ii), d(iii)

59. Which among the following pair of elements are isotones?
(1) $^{36}_{16}S$ and $^{39}_{19}K$
(2) $^{37}_{17}Cl$ and $^{13}_{6}C$
(3) $^{14}_{7}N$ and $^{16}_{8}O$
(4) $^{2}_{1}H$ and $^{4}_{2}He$

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60. Match column-I with column-II.

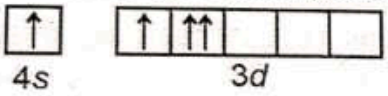
	Column-I (Orbital)		Column-II (Node)
a.	$3d_{z^2}$	(i)	No nodal plane, 1 radial node
b.	$2p_z$	(ii)	No nodal plane, zero radial node
c.	$2s$	(iii)	yz and xz nodal plane, one radial node
d.	$4d_{xy}$	(iv)	xy nodal plane, zero radial node

In the light of above statements, choose the correct option.

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(iv), c(i), d(iii)

(3) a(iii), b(i), c(ii), d(iv) (4) a(iv), b(iii), c(i), d(ii)

61. Consider the following electronic configuration



The rule(s) violated is/are

a. Pauli exclusion principle

b. Hund's rule of maximum multiplicity

c. Aufbau principle

Choose the correct option.

(1) b and c only (2) c only

(3) a and b only (4) a, b and c

62. Given below are two statements:

Statement I: n, ℓ, m_ℓ and m_s arise as a natural consequence in the solution of Schrodinger equation.

Statement II: The wave function is a mathematical function whose value depends upon the coordinates of the electron in the atom and does not carry any physical meaning.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are true

(2) Statement I is false but statement II is true

(3) Statement I is true but statement II is false

(4) Both statement I and statement II are false

63. If the kinetic energy of He is four times the kinetic energy of Ne. The ratio of the de-Broglie wavelength of Ne to that of the de-Broglie wavelength of He is

(1) $\sqrt{2} : \sqrt{7}$

(2) $\sqrt{5} : \sqrt{2}$

(3) $2 : \sqrt{5}$

(4) 2 : 1

64. A ball has a mass of 30 g and a speed of 50 m/s. If the uncertainty in the speed is 3% then the uncertainty in position will be

(1) 1.91×10^{-33} m

(2) 3.91×10^{-33} m

(3) 2.71×10^{-33} m

(4) 1.17×10^{-33} m

65. The energy of an electron in the first Bohr orbit of H-atom is –13.6 eV. The possible energy values of the excited states for electron in He^+ ion is

a. – 13.6 eV b. – 6.04 eV

c. – 3.4 eV

(1) a and b only

(2) a and c only

(3) b and c only

(4) a, b and c

66. The wave number for the shortest wavelength transition in Paschen series of atomic hydrogen is

(1) 12186.3 cm^{-1}

(2) 17214.3 cm^{-1}

(3) 13014.2 cm^{-1}

(4) 15912.1 cm^{-1}

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67. The work function of some metals is listed below. The number of metals which will show photoelectric effect when light of 248 nm wavelength falls on the metal is

Metal	ϕ (eV)
Li	2.42
Na	2.3
K	2.25
Mg	3.7
Cu	4.8
Ag	4.3
Pt	6.3

(1) 7 (2) 5
(3) 6 (4) 4

68. Which among the following statements is/are true about canal rays?

a. The behaviour of the particles of canal rays in the magnetic or electric field is opposite to that observed for cathode rays.

b. Canal rays consist the positively charged gaseous ions.

c. The charge to mass ratio of the particles of canal rays depend on the gas from which these originate.

Choose the correct option.

(1) a and b only (2) b and c only
(3) a and c only (4) a, b and c

69. 1.03 g mixture of Na_2CO_3 and $CaCO_3$ requires 20 mL, 1 N HCl for complete neutralisation. Percentage of $CaCO_3$ in the mixture is

(1) 23% (2) 35%
(3) 58.72% (4) 48.54%

70. Mass of 11.2 L CO_2 at STP will be

(1) 44 g (2) 22 g
(3) 11.2 g (4) 22.4 g

71. An organic chloro compound contains 71.7% Cl, 4.04% H and rest carbon. Empirical formula of the compound will be

(1) CH_2Cl (2) $CHCl_3$
(3) CH_2Cl_2 (4) CH_3Cl

72. A 3 L gaseous mixture of butane and propane gives 10 L CO_2 on complete combustion at constant temperature and pressure. Volume of butane and propane in initial mixture were respectively

(1) 1.5 L and 1.5 L (2) 2 L and 1 L
(3) 1 L and 2 L (4) 2.5 L and 0.5 L

73. 9 mL of pure dry O_2 is subjected to silent electric discharge, if 10% of O_2 is converted to O_3 then final volume of gaseous mixture will be

(1) 8.7 mL (2) 8 mL
(3) 9.3 mL (4) 9.1 mL

74. A mixture of 4.6 g formic acid and 9 g oxalic acid is treated with conc. H_2SO_4, where conc. H_2SO_4 acts as a dehydrating agent. The evolved gaseous mixture is passed through KOH pellets. Weight (in gram) of the remaining gaseous product at STP will be

(1) 2.8 (2) 10
(3) 5.6 (4) 11.2

75. A crystalline salt $Na_2SO_4 \cdot xH_2O$ on heating loses 55.9% of its mass and becomes anhydrous. The formula of crystalline salt is

(1) $Na_2SO_4 \cdot 10H_2O$ (2) $Na_2SO_4 \cdot 2H_2O$
(3) $Na_2SO_4 \cdot 7H_2O$ (4) $Na_2SO_4 \cdot 5H_2O$

76. If 2 g H_2 is mixed with 14 g N_2 to produce NH_3 in a closed vessel. Consider the following statements.

a. Mass of reaction mixture will be 16 g at the end of the reaction.

b. 16 g NH_3 will be produced.

c. 666.67 ml of 1 M HCl solution is required to completely neutralise the gaseous mixture obtained after the completion of the reaction.

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d. Volume of NH_3 evolved at STP will be nearly 21.08 L.

e. 4.67 g N_2 will be left unreacted

Correct statements are

(1) a, c and e only (2) b and d only

(3) a, b, c and d only (4) c and d only

77. Match prefix in column-I with their numerical value in column-II.

	Column-I		Column-II
a.	Femto	(i)	10^{15}
b.	Micro	(ii)	10^{12}
c.	Tera	(iii)	10^{-6}
d.	Peta	(iv)	10^{-15}

Correct match is

(1) a(i), b(iii), c(ii), d(iv) (2) a(iv), b(ii), c(iii), d(i)

(3) a(ii), b(iii), c(i), d(iv) (4) a(iv), b(iii), c(ii), d(i)

78. Molarity of sugar ($C_{12}H_{22}O_{11}$) solution obtained on dissolving 3.42 g sugar in 200 ml solution will be

(1) 0.5 M (2) 0.1 M

(3) 0.01 M (4) 0.05 M

79. Given below are two statements, one is **Assertion (A)** other is **Reason (R)**

Assertion (A): 1 L of both N_2 and H_2 at 2 atm pressure and 100°C temperature contain equal number of molecules.

Reason (R): Equal volume contains equal number of molecules at same temperature and pressure.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

80. Among the following choose ones with equal number of atoms

a. 2.24 L of SO_3 at STP

b. 4.8 g of ozone

c. 4.4 g of N_2O

d. 0.2 mol of NH_3

e. 0.2 N_A molecules of water

(1) a, b and c only

(2) b and c only

(3) b, c and d only

(4) a, c and d only

81. Minimum mass of barium chloride required to produce 11.65 g barium sulphate on reaction with 9.1 g sodium sulphate will be (Atomic mass of Ba = 137 u)

(1) 20.8 g

(2) 10.4 g

(3) 41.6 g

(4) 5.7 g

82. Minimum molar mass of a macromolecule that contains 0.01% of Ca by mass is

(1) 400 kg (2) 10^4 u

(3) 10 kg (4) 4×10^5 u

83. Consider the following statements for 0.15 g CO_3^{2-} ions (take Avogadro number $N_A = 6 \times 10^{23}$)

a. Number of ions are 1.5×10^{21}

b. Total charge is nearly 965 coulomb

c. Total number of moles of electrons is 0.08

d. Mass of carbon is 0.03 g

Correct statements are

(1) a, b and c only (2) a, b and d only

(3) a, b, c and d (4) a, c and d only

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84. Consider the following statements:

a. Cisplatin is used in the treatment of cancer.

b. Both 2.00 and 200 have three significant figures.

c. SO_2 and SO_3 follows law of multiple proportions.

d. Bronze is a heterogenous mixture.

Correct statement(s) is/are

(1) c only

(2) a, b and c only

(3) a, b, c and d

(4) a and c only

85. Concentrated aqueous H_2SO_4 is 98% H_2SO_4 by mass and has a density of 1.8 g/mL. Volume of the acid required to make one litre of 0.5 M H_2SO_4 is

(1) 55.5 mL

(2) 27.78 mL

(3) 5.55 mL

(4) 12.1 mL

86. An element X has following isotopic compositions ^{100}X : 90%, ^{99}X : 8% and ^{102}X : 2%.

The weighted average atomic mass of the naturally occurring element X is

(1) 99.96 u

(2) 100.04 u

(3) 100 u

(4) 100.33 u

87. 10 g mixture of NaCl and Na_2CO_3 is heated with dilute HCl and 2.2 g CO_2 is evolved. Mass percentage of NaCl in the mixture is

(1) 53%

(2) 67%

(3) 47%

(4) 33%

88. Match the SI units in column-I with base physical quantity in column-II.

	Column-I		Column-II
a.	Mole	(i)	Mass
b.	Kilogram	(ii)	Thermodynamic temperature
c.	Kelvin	(iii)	Amount of substance
d.	Second	(iv)	Time

Correct match is

(1) a(iii), b(i), c(ii), d(iv)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(i), b(iii), c(iv), d(ii)

89. Consider the following mixtures

a. aq. NaCl b. Air

c. Milk d. Iron filings

Heterogeneous mixture(s) is/are

(1) a and b only

(2) d only

(3) a, b and c only

(4) c only

90. Number of molecules of CO left if 1.4 mg CO is removed from 2.24 ml of CO gas at STP will be

(1) 1.5×10^{22}

(2) 4.5×10^{19}

(3) 3×10^{19}

(4) 1.5×10^{19}

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[BOTANY]

91. Carnivora includes lower taxa like
(1) Felidae and Canidae
(2) Mammalia and Hominidae
(3) Primata and Insecta
(4) Felidae and Diptera

92. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Family	(i)	Comprises a group of related species
b.	Order	(ii)	Is a group of related genera with still less number of similarities as compared to genus and species
c.	Genus	(iii)	Includes related taxa of Primata and Diptera etc.
d.	Class	(iv)	It's an assemblage of certain taxa like Solanaceae, Convolvulaceae

(1) a-iv, b-ii, c-i, d-iii (2) a-ii, b-iv, c-i, d-iii
(3) a-iii, b-i, c-ii, d-iv (4) a-ii, b-i, c-iv, d-iii

93. Select the **incorrectly** matched pair.
(1) Species — *leo*
(2) Genus — *Solanum*
(3) Family — Sapindales
(4) Class — Monocotyledonae

94. Read the following statements and select the **correct** option.
Statement A: Chlorophyll pigments are present in the thylakoids.
Statement B: An elaborate network of filamentous proteinaceous structures consisting of microtubules, microfilaments and intermediate filaments present in the cytoplasm collectively referred to as cytoskeleton of eukaryotes.
(1) Both the statements are correct
(2) Both the statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

95. Yeast cell can progress through the cell cycle in only about
(1) 24 hours (2) 24 minutes
(3) 90 minutes (4) 90 hours

96. Mitotic division occurs in the
a. Root apical meristem b. Lateral cambium
c. Embryo d. Gamete
Choose the **correct** ones
(1) a, b and c only (2) b, c and d only
(3) c and d only (4) b and d only.

97. Read the following **assertion (A)** and **reason (R)** statements and select the correct option.
Assertion (A): Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing higher plants.
Reason (R): In higher plants, during gametogenesis meiosis reduces the chromosome number by half and produces haploid gametes.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

98. Shortening of microtubules attached to kinetochores occurs during
(1) Metaphase I (2) Anaphase II
(3) Telophase (4) Prophase I

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99. Crossing over occurs between
(1) Non-homologous chromosomes having sister chromatids
(2) Sister chromatids of homologous chromosomes
(3) Non-sister chromatids of homologous chromosomes
(4) Non-sister chromatids of non-homologous chromosomes

100. The liquid endosperm in coconut is a result of
(1) Karyokinesis not followed by cytokinesis
(2) Cytokinesis followed by karyokinesis
(3) Karyokinesis followed by cytokinesis
(4) Cytokinesis not followed by karyokinesis

101. In plants, the formation of the new cell wall begins with the formation of a simple precursor. In a mature plant cell this structure is mainly comprised of
(1) Calcium carbonate (2) Calcium pectate
(3) Mannans (4) Cellulose

102. In an animal cell, the end of M phase starts with
(1) The appearance of a furrow in the plasma membrane.
(2) Reformation of nucleolus, Golgi complex and ER.
(3) Equal division of mitochondria and chloroplast in daughter cells.
(4) Formation of cell wall in the centre of the cell.

103. At the beginning of the final stage of karyokinesis,
(1) Centromeres split and chromatids separate
(2) The X-shape of the chromosome is clearly observed
(3) The chromosomes that have reached their respective poles decondense and lose their individuality
(4) The chromosomal material becomes untangled during the process of chromatin condensation.

104. If the initial amount of DNA is denoted as 2C and the number of chromosomes are 16 in onion root tip cells, then at metaphase stage the cell will have
(1) 4C amount of DNA and 16 chromosomes
(2) 4C amount of DNA and 32 chromosomes
(3) 2C amount of DNA and 32 chromosomes
(4) 2C amount of DNA and 16 chromosomes

105. Select the most appropriate sequence of a cell cycle events observed in the cell that is undergoing mitosis.
a. Cells will exhibit a stage in which they remain metabolically active but no longer proliferate.
b. Chromosomes are spread through the cytoplasm of the cell. (3)
c. Centrosome undergoes duplication in the cytoplasm of animal cells. (2)
d. Daughter chromosomes of the future daughter nuclei begin their migration towards the two opposite poles.
e. Chromosomes are moved to spindle equator and get aligned along metaphase plate.
f. Cell is metabolically active and continuously grows but does not replicate its DNA, though most of the organelles are duplicated. (1)
(1) f → c → b → d → e
(2) f → c → b → e → d
(3) a → f → a → c → b → e → d
(4) a → f → b → c → e → d

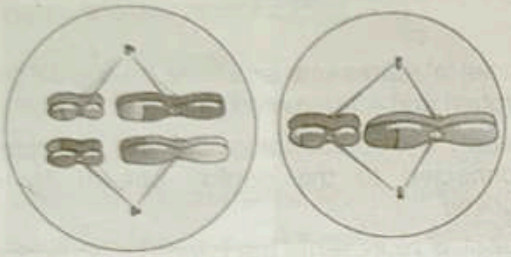
106. The beginning of diplotene is recognised by the
(1) Formation of protein complex that holds the homologous chromosomes together.
(2) Dissolution of synaptonemal complex and the tendency of the recombined homologous chromosomes of bivalents to separate from each other except at crossover site.
(3) Terminalisation of chiasmata and fully condensed chromosomes.
(4) Disappearance of nucleolus and nuclear envelope.

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107. Select the **incorrect** statement.
(1) Prophase of the first meiotic division is typically longer and more complex when compared to prophase of mitosis.
(2) During zygotene stage, chromosomes start pairing together and this process of association is called synapsis.
(3) Bivalent or a tetrad is more clearly visible in the shortest phase of prophase I.
(4) The compaction of chromosomes continues throughout the prophase I.

108. The below given cells are in a particular stage of cell cycle. Which of the given statement is **correct** for them?



(a) (b)

(1) The stage 'b' will undergo next stage with the aid of anaphase promoting complex.
(2) The stage 'b' brings out variation in the gametes as crossing over occurs during this phase.
(3) Stage 'a' represents anaphase I while stage 'b' represents metaphase I.
(4) Stage 'a' is seen in haploid cells while 'b' occurs in diploid cells.

109. Double membrane bound organelle that bears ribosome on one of its membrane is
(1) Endoplasmic reticulum
(2) Nucleus
(3) Mitochondria
(4) Chloroplast

110. Respiration in prokaryotes is carried out by
(1) Mitochondria (2) Microbodies
(3) Polysome (4) Mesosome

111. Synthesis of lipoproteins occurs in
(1) Lysosomes
(2) Ribosome
(3) Endoplasmic reticulum
(4) Vacuole

112. The interphase nucleus has
(1) Highly condensed chromatin known as chromosomes
(2) Highly extended and elaborate nucleoprotein fibres
(3) Twice the number of chromosomes than in the nucleus after mitosis
(4) Chromatin, basic proteins and histones but not RNA

113. Select the **incorrectly** matched pair.
(1) Endoplasmic reticulum — Network or reticulum of tiny tubular structures scattered in the cytoplasm
(2) Cytoskeleton — Are involved in many functions such as mechanical support, motility, maintenance of the shape of cell
(3) Thylakoids — Are arranged in stacks like piles of coins called grana
(4) Kinetochores — Primary constrictions that hold two chromatids of a chromosome

114. The eukaryotic ribosome has two subunits
(1) 60S and 40S that together forms a 70S ribosome
(2) 40S and 30S that together forms an 80S ribosome
(3) 60S and 40S that together forms an 80S ribosome
(4) 50S and 30S that together form an 80S ribosome

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115. Match the following columns and select the correct option.

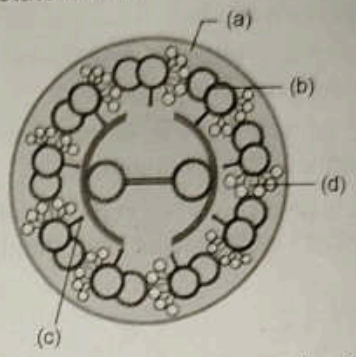
	Column I		Column II
a.	George Palade	(i)	He first saw and described a live cell.
b.	Camillo Golgi	(ii)	He was first to observe densely stained reticular structures near the nucleus.
c.	Robert Brown	(iii)	He was the first to observe a dense particle whose composition is now known to be only RNA and proteins.
d.	Antonie Von Leeuwenhoek	(iv)	He was the first one to describe the nucleus.

(1) a(ii), b(iii), c(i), d(iv) (2) a(iii), b(ii), c(iv), d(i)
(3) a(iii), b(iv), c(ii), d(i) (4) a(iv), b(ii), c(iii), d(i)

116. The organelle that is present in animal cell but absent in higher plant cell shows all of the following features, **except**
(1) It contains two cylindrical structures that are surrounded by amorphous pericentriolar material.
(2) This structure is made up of nine evenly spaced peripheral fibrils of tubulin protein.
(3) The central part of the proximal region of this cylindrical structures is called hub.
(4) The axoneme of this structure has nine doublets of radially arranged peripheral microfilaments.

117. Reserve material in prokaryotic cells are stored in the cytoplasm in the form of
(1) Single unit membrane bound system that lie free in the cytoplasm.
(2) Non-membrane bound organelle exclusively made up of proteins.
(3) Phosphate, cyanophycean and glycogen granules in the cytoplasm.
(4) Contractile vacuoles as found in blue-green, purple and green photosynthetic bacteria.

118. Identify the below given structure and choose the **correct** statements.



(1) Label 'a' represents axoneme having 52 percent protein and 40 percent lipids.
(2) Label 'b' represents peripheral microtubules connected to the central sheath by radial spokes.
(3) Label 'c' represents the interdoublet bridge that connect the central microtubule with the peripheral ones.
(4) Label 'd' represents the radial spokes made of tubulin protein.

119. Which of the below given features are **true** for endomembrane system?
(1) The members are only observed in those cells that are actively involved in protein synthesis and secretion.
(2) All the members perform the function of packaging materials, to be secreted *outside* of the cell.
(3) One of the members acts as an important structure for osmoregulation and excretion in *Amoeba*.
(4) Some of the members are non-membrane bound granular structures exclusively composed of rRNA and proteins.

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120. Select the **odd** one
(1) Cell wall helps in cell-to-cell interaction and provides barrier to undesirable macromolecules.
(2) Algae have cell membrane made up of cellulose, galactose, mannans and minerals like calcium carbonate.
(3) In a young plant cell, the primary cell wall is capable of growth.
(4) The secondary wall in plant cell is formed on the inner side towards membrane side of the cell.

121. Scientific name of housefly is **correctly** printed as
(1) Musca Domestica
(2) *Musca* domestica
(3) *Musca domestica*
(4) Musca domestica

122. Select the **mismatched** pair from the following:
(1) Angiospermae — Division
(2) Poales — Order
(3) Dicotyledonae — Phylum
(4) Muscidae — Family

123. Read the following statements A and B and select the correct option.
Statement A: The number of similar characteristics goes on increasing from species to kingdom.
Statement B: Higher the category, greater is the difficulty of determining the relationship to other taxa at same level.
(1) Both statements are correct
(2) Only statement B is correct
(3) Only statement A is correct
(4) Both statements are incorrect

124. The significance of meiosis lies in
a. Reduction of the diploid number of chromosomes to haploid.
b. Maintaining ploidy during sexual reproduction.
c. Production of genetic variability in the population of a species.
(1) Only a and b
(2) Only b and c
(3) Only c
(4) All a, b and c

125. In the somatic cell cycle
(1) A short interphase is followed by a long mitotic phase
(2) G_2 phase immediately follows the mitotic phase
(3) In G_1 phase, the DNA content is double than the amount of DNA present in the parent cell
(4) DNA replication and histone protein synthesis occur in the synthesis phase

126. Colchicine inhibits polymerisation of microtubules hence there occurs no spindle formation. This will immediately inhibit the ____.
(1) Prophase (2) Metaphase
(3) Anaphase (4) Telophase

127. Select the **incorrect** statement.
(1) During anaphase, the centromere of each chromosome is towards the pole and their arms trail behind.
(2) Kinetochore serves as the site for spindle fibre attachment.
(3) Phenomenon of bringing chromosomes at pole is called congression.
(4) In the S and G_2 phases, the new DNA molecules formed are not distinct but intertwined.

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128. Meiosis II and mitosis differ from each other in several features. Which of the following option **correctly** differentiates them?

(1) Meiosis II occurs only in a haploid cell while mitosis in a diploid or a haploid cell

(2) Meiosis II is not immediately followed by DNA replication but mitosis is followed by DNA replication

(3) In meiosis II, the two chromatids of a chromosome are always identical but in mitosis they are not

(4) The daughter cells formed after meiosis II are genetically similar to each other but in mitosis, they are not

129. "A student observed a bacterium dividing by binary fission."
How many features of living beings are shown by that bacterium through its division?

(1) 2 characteristic features

(2) Only 1 characteristic feature

(3) 2 defining features

(4) Only 1 defining feature

130. Read the following **assertion (A)** and **reason (R)** statements and select the **correct** option.

Assertion (A): All cells arise from pre-existing cells.

Reason (R): All living organisms are composed of cells and products of cells.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

131. RNA that is not found in the eukaryotic 80S ribosome is

(1) 5.8S (2) 16S

(3) 5S (4) 28S

132. Which one of the following pairs is **correctly** matched?

(1)	Microbody	Participate in the process of photosynthesis.
(2)	Lysosomes	Involved in synthesizing amino acids.
(3)	Endoplasmic reticulum	Plays a role in the formation of lysosomes.
(4)	Centrosome	Provide enzymes required in the digestive process.

133. The presence of DNA in mitochondria and chloroplast supports the hypothesis that

(1) Both are bacterial endosymbionts of the cell

(2) Glycolysis occurs in both the organelles

(3) ATP is produced in mitochondria and chloroplast

(4) Both undergo mitosis and meiosis independent of nucleus

134. Identify the organelles of the endomembrane system on the basis of given features.

A. Associated with muscle contraction by release and uptake of Ca^{+2} ions.

B. Secretes mucilage for the lubrication of root tip.

C. Shows polymorphism.

	A	B	C
(1)	Golgi bodies	Vacuoles	Lysosomes
(2)	SER	Golgi bodies	Lysosomes
(3)	RER	Plastids	Golgi bodies
(4)	SER	Lysosome	Golgi bodies

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135. Which of the following statements is **incorrect** for the plasma membrane?
(1) The major lipids found in the membrane are phospholipids that are arranged in a bilayer.
(2) In human RBCs, it is completely impermeable.
(3) Integral proteins are partially or totally buried in the membrane.
(4) The fluid nature of the membrane is important for cell growth.

[ZOOLOGY]

136. In a neuron, the cell body containing the nucleus is also called
(1) Dendrite
(2) Cyton
(3) Axon hillock
(4) Astrocyte

137. Which biological fact explains as to why neural tissue membranes possess complex structural lipids in abundance?
(1) The continuous cellular migration exhibited by mature neurons of CNS for contraction
(2) The use of only glycerol as the primary oxidative fuel during depolarization
(3) The physiological requirement for efficient insulation of nerve fibres
(4) The storage of metabolic wastes

138. Select the correct match w.r.t. elements and their % weight in human body.
(1) Hydrogen - 0.14%
(2) Carbon - 18.5%
(3) Oxygen - 46.6%
(4) Nitrogen - 9.5%

139. How many of the following is/are incorrect?
a. Lipids do not contain oxygen in their structure.
b. Lipids contain a high ratio of hydrogen to oxygen atoms.
c. Carbohydrates always exhibit a ringed structure.
d. Lipids and carbohydrates are always non-polymeric.

Select the **correct** option.
(1) Four
(2) Two
(3) Three
(4) One

140. A tissue containing matrix rich in chondroitin sulphate and cells enclosed in lacunae, but no blood vessels is exemplified by
(1) Areolar tissue
(2) Fluid connective tissue
(3) Skeletal connective tissue
(4) Dense connective tissue

141. Which of the following options represents the correct statement about tendon or ligament?
(1) Tendon connects bone to bone and is elastic in nature.
(2) Ligaments connect bone to muscle and are inelastic in nature.
(3) Tendons are categorised under dense irregular connective tissues.
(4) Ligaments are categorised under dense regular connective tissues.

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142. Consider the following statements:

Statement A: Cells of epithelial tissue are held together with little intercellular material.

Statement B: In all animals, tissues have specialized junctions that provide both structural and functional links between individual cells.

Choose the correct option.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is incorrect
(4) Only statement B is incorrect

143. **Assertion (A):** Cellulose is indigestible in humans.

Reason (R): Cellulose is a storage polysaccharide in plants.

In the light of above statements, select the correct option.

(1) Both (A) and (R) are true, and (R) explains (A) correctly
(2) Both (A) and (R) are true, but (R) does not explain (A) correctly
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

144. A biomolecule is insoluble in water, forms bilayer in aqueous medium and contains glycerol along with a phosphate moiety. It is most likely

(1) Palmitic acid (2) Cholesterol
(3) Phospholipid (4) Polysaccharide

145. Choose the **incorrect** statement.

(1) K_m value is inversely related to affinity between a substrate and an enzyme
(2) Competitive inhibitors increase apparent K_m but do not change V_{max}
(3) Amino acids exhibit the zwitterionic form
(4) Enzymes are not substrate specific

146. **Assertion (A):** Proteins exhibit a wide range of biological functions.

Reason (R): Tertiary structure of protein, which is essential for its biological activity, can get denatured without even breaking the peptide bonds.

In the light of above statements, select the correct option.

(1) Both (A) and (R) are true, but (R) does not explain (A) correctly
(2) Both (A) and (R) are true, and (R) explains (A) correctly
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

147. The feature(s) common to both ligaments and cartilage is/are

a. Presence of chondrocytes
b. Presence of collagen fibres
c. Absence of osteocytes
d. Presence of matrix

Select **correct** option.

(1) a, b, c and d (2) b, c and d only
(3) c and d only (4) d only

148. In a histological microscopic slide, the tissue shows branched fibres, uninucleated cells and intercalated discs. The tissue is most likely

(1) Skeletal muscle (2) Smooth muscle
(3) Cardiac muscle (4) Areolar tissue

149. Select the **incorrect** match.

(1) Mast cells – Histamine secretion
(2) Plasma cells – Antibody production
(3) Goblet cells of alimentary canal – Multicellular glands
(4) Fibroblasts – Fibre formation

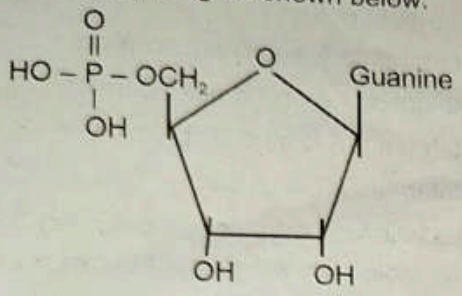
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150. Choose the epithelial tissue that is present in the initial bronchioles.

(1) Transitional epithelium
(2) Ciliated epithelium
(3) Compound epithelium
(4) Simple squamous epithelium

151. Consider the figure shown below:



Which of the following statements is **not** true w.r.t. the above shown figure?

(1) It represents a nucleotide that lacks 2' deoxyribose sugar
(2) The given molecule has 3 heterocyclic rings in its structure
(3) Glycosidic bond is absent in this molecule
(4) This molecule is not the component of DNA

152. While primary metabolites have __A__ functions, we do not at the moment __B__ the physiological roles or functions of __C__ secondary metabolites in host organisms. Select the option to fill the blanks A, B, and C correctly.

	A	B	C
(1)	Unknown	Know	All the
(2)	Identifiable	Understand	All the
(3)	Undefined	Identify	All the
(4)	Unknown	Define	All the

153. A tissue section shows the following features:

a. Matrix heavily mineralized.
b. Communication between cells through canaliculi.
c. Surrounded by a tough sheath, called periosteum.

Identify the tissue based on the above given features and select the correct option.

(1) Cartilage
(2) Bone
(3) Dense irregular connective tissue
(4) Tendon

154. Consider the following statements:

a. Tight junctions mainly facilitate ionic communication.
b. Gap junctions help cells to communicate chemically and electrically.
c. Adhering junctions are specialised for rapid impulse conduction.
d. Communication junctions prevent passage of substances between cells (cytoplasm).

Which of the above is/are **not** true?

(1) a, b, c and d
(2) a, c and d only
(3) a, b and d only
(4) b, c and d only

155. How does essential metal ion co-factors interact within the active catalytic pocket of an enzyme to accelerate the transformation of a substrate?

(1) By inducing permanent genetic mutation in the amino acid sequence of the active site
(2) By denaturing the local secondary configuration of the enzyme to expose polar amino acids
(3) By forming covalent peptide cross-links that permanently anchor the substrate into the tertiary matrix
(4) By forming coordination bonds with the side chains at the active site and simultaneously with the substrate molecule

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156. A catalytically active conjugated enzyme is composed of protein and non-protein components. What is the precise terminology for the isolated protein part of the enzyme that remains inactive on its own?

(1) Prosthetic group (2) Apoenzyme
(3) Holoenzyme (4) Co-enzyme

157. A student observed the following features while studying an animal tissue:

(a) Abundant ground substance
(b) Loosely arranged fibres
(c) Macrophages, fibroblasts and mast cells present in its matrix

Based on these features, the tissue is most likely involved in

(1) Rapid conduction of impulses
(2) Voluntary contraction
(3) Binding skin to muscle
(4) Storage of calcium phosphate

158. A tissue taken from the wall of the intestine shows presence of spindle-shaped cells, absence of striations and centrally located nuclei.

This tissue

(1) Is voluntary in nature
(2) Is also found in the wall of aorta
(3) Exhibits branching
(4) Contains only elastin and myosin

159. Choose the connective tissue that has a fluid matrix without fibres and possesses cells that can show immune functions.

(1) Cartilage (2) Adipose tissue
(3) Blood (4) Bone

160. Consider the following statements:

Statement A: Living organisms have a number of carbon compounds and all of them contain heterocyclic rings.

Statement B: When a hexose sugar and a phosphate group are attached to a nitrogenous base, it forms a nucleoside.

Select the **correct** option.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is incorrect
(4) Only statement A is correct

161. Which of the following organic compound(s) do(es) not have/has molecular weights in the range of ten thousand daltons and above?

a. Proteins
b. Nucleic acids
c. Polysaccharides
d. Lipids

Choose the **correct** option.

(1) a, b, c and d
(2) b, c and d only
(3) c and d only
(4) d only

162. How many of the compounds mentioned in the box given below are secondary metabolites which are useful to humans?

Morphine, Amino acids, Antibiotics, Lecithin, Spices

Select the **correct** option.

(1) Five (2) Four
(3) Three (4) Two

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163. Upon complete hydrolysis, a peptide chain yields only two molecules of valine and two molecules of serine. This indicates that

(1) This chain contains only four peptide bonds

(2) This peptide chain is an oligopeptide

(3) The molecular weight of the peptide must be above 3000 Da

(4) Both valine and serine contain $-CH_2OH$ as the 'R' group

164. Which of the following statements is/are **incorrect** w.r.t. nucleic acids?

(a) Uracil is absent in RNA.

(b) DNA strands are anti-parallel to each other.

(c) Phosphodiester bonds join adjacent nucleotides of opposite strands of a dsDNA molecule.

(d) Substituted purines are double ring nitrogenous bases.

Choose the **correct** option.

(1) (a) and (b)

(2) (c) and (d)

(3) (a) and (c)

(4) (b) and (d)

165. How does dense regular connective tissue differs from dense irregular connective tissue?

(1) Former lacks collagen fibres

(2) In former, collagen fibres are oriented in parallel bundles, whereas collagen fibres are oriented randomly in latter

(3) Former is highly vascularized, whereas latter is completely avascular

(4) Only former forms the subcutaneous layer beneath the skin epithelium

166. Select the layer of connective tissue that is present directly beneath the skin, serves as a support framework for epithelium and contains loose arrangement of fibres in its matrix.

(1) Epithelial tissue

(2) Dense regular connective tissue

(3) Areolar tissue

(4) Skeletal connective tissue

167. Which of the following descriptions is accurate about compound epithelium?

(1) It relies entirely on inner lining of blood vessels penetrating into apical layer of epithelium

(2) It consists of multilayered cells with tight junctions that allow rapid movement of ions across the tissue

(3) It is exclusively composed of modified columnar cells containing large number of goblet cells

(4) It is multilayered, has a limited role in secretion and primarily provide protection against chemical and mechanical stresses

168. Bone cells, called 'X', are embedded inside rigid concentric matrix layers. 'X' are present in the fluid-filled spaces called 'Y'.

Choose the option that correctly represents 'X' and 'Y', respectively.

(1) Osteoblasts; Chondrin

(2) Osteocytes; Lacunae

(3) Fibroblasts; Lamellae

(4) Marrow cells; Lacunae

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169. How do enzymes alter the thermodynamic path of a biochemical reaction?
(1) By raising the activation energy barrier needed for the reaction to move in the forward direction
(2) By changing the absolute potential energy value of both the substrate and the final products
(3) By lowering the activation energy barrier, making it easier for the substrate to reach the unstable transition state
(4) By absorbing external thermal energy to decrease the velocity of the reaction

170. Complete the analogy by selecting the correct option.
Trypsin, Insulin : Heteropolymers :: GLUT-4, Collagen : _____ .
(1) Homopolysaccharides
(2) Heteropolysaccharides
(3) Homopolymers
(4) Heteropolymers

171. While the classic axiom states that "all enzymes are proteins", but there are clear exceptions to this. Which of the following molecules breaks this rule?
(1) DNA polymerase
(2) Ribozyme
(3) Nuclease
(4) Carbonic anhydrase

172. Match List-I with List-II.

	List-I		List-II
a.	Maltose	(i)	Absence of complex helices
b.	Glycogen	(ii)	Polymer of N-acetylglucosamine
c.	Cellulose	(iii)	Disaccharide made up of glucose
d.	Chitin	(iv)	Branched storage polysaccharide

Choose the correct answer from the options given below.
(1) a(iv), b(iii), c(ii), d(i) (2) a(i), b(ii), c(iii), d(iv)
(3) a(ii), b(i), c(iv), d(iii) (4) a(iii), b(iv), c(i), d(ii)

173. A researcher heats an enzyme at high temperature and observes loss of catalytic activity. Even after cooling, the activity is not restored. Which of the following events most likely led to inactivity of the enzyme?
(1) Hydrolysis of ester bonds of the primary protein structure
(2) Breaking of glycosidic linkage between individual monomeric units
(3) Permanent disruption of the tertiary structure of the protein
(4) Conversion of amino acids into glycogen

174. Which specific structural component of the human neural system contributes more than half the total volume of neural tissue, and also has protective and supportive role?
(1) Myelinated nerve fibres
(2) Neuroglial cells
(3) Nissl's granules
(4) Multipolar neurons

175. Choose the amino acid that possesses a cyclic structure with a straight side chain bearing carboxylic and amino groups.
(1) Phenylalanine (2) Alanine
(3) Cysteine (4) Methionine

176. Which of the following epithelial tissues is specialized for minimizing resistance for diffusion of gases?
(1) Stratified squamous epithelium
(2) Brush border cuboidal epithelium
(3) Simple squamous epithelium
(4) Ciliated columnar epithelium

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177. The ciliated simple epithelium in humans is functionally indispensable due to its ability to move particles or mucus in a specific direction. Where is it primarily located in the human body?

(1) Tubular parts of nephrons in kidneys

(2) Ducts of salivary glands

(3) Intestinal mucosa

(4) Inner layer of the wall of the structure where zygote is formed

178. Select the incorrect statement among the following.

(1) Physicists conjure up the four dimensional views of molecular structures, while biologists describe the protein structure at only three levels.

(2) A protein thread does not exist throughout as an extended rigid rod.

(3) In proteins, only right handed helices are observed.

(4) An α-helix of protein has intramolecular hydrogen bonding.

179. Select the odd one w.r.t. sugar composition.

(1) Glycogen

(2) Starch

(3) Inulin

(4) Cellulose

180. To perform chemical analysis, the living tissue is grinded in

(1) $ClCH_2COOH$, using only a pestle

(2) Trichloroacetic acid, using a mortar and a pestle

(3) $Cl_2HCCOOH$, using cheese cloth

(4) Trihydroxypropane, using a pestle

❑ ❑ ❑

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